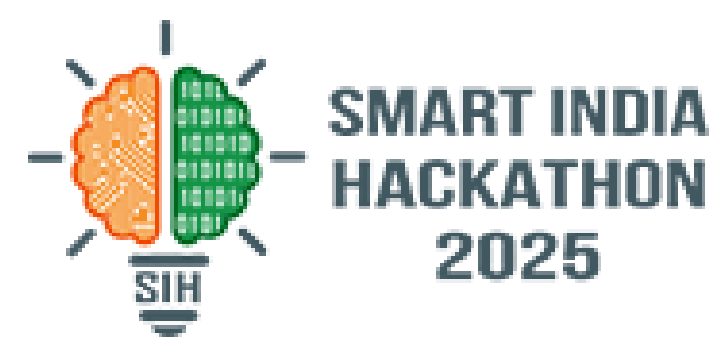


SMART INDIA HACKATHON 2025



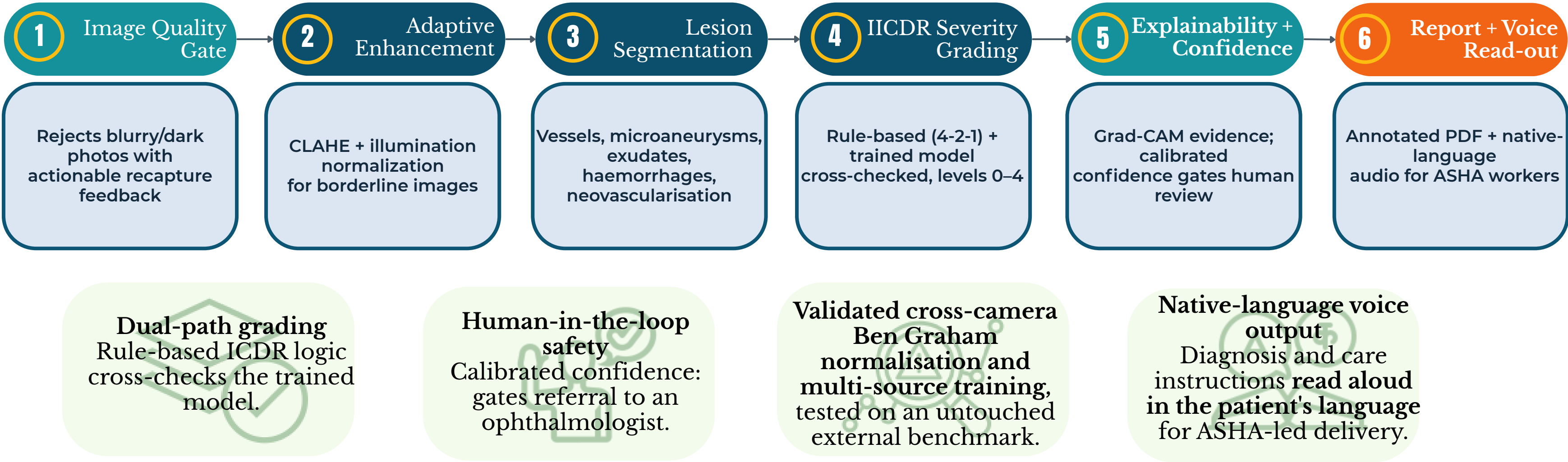
- Problem Statement ID – SIH26038
- Problem Statement Title- Explainable AI for Diabetic Retinopathy Screening in Rural India
- Theme- MedTech / BioTech / HealthTech
- PS Category- Software
- Team Name (Registered on portal)- Veyronix



DRISHTI

Explainable AI Diabetic Retinopathy Screening for Rural India

Kinesio-style remote screening pipeline: A MATLAB + Python pipeline that **grades diabetic retinopathy severity from a fundus photo**, **explains its reasoning** with lesion-level evidence and Grad-CAM, **routes uncertain cases to a human**, and **models district-level rollout capacity** — so a single ophthalmologist can safely screen 100,000+ patients/year through ASHA workers and PHCs.



TECHNICAL APPROACH

Technology Stack

MATLAB (for deployment)

1. Image Processing Toolbox
2. Computer Vision Toolbox
3. Deep Learning Toolbox (+ ONNX Converter)
4. Simulink
5. Statistics & ML Toolbox

Python (for training)

1. PyTorch + timm (EfficientNet-B3)
2. Albumentations
3. ONNX Runtime
4. FastAPI gateway

Datasets

1. APTOS 2019 (~3,662 imgs)
2. IDRiD (Indian, segmentation + grading)
3. Messidor-2 (external benchmark, 1,744 imgs)

Train fast in PyTorch → export ONNX → import into MATLAB (importNetworkFromONNX) for inference + Grad-CAM — a deliberate hybrid, not a workaround: fastest iteration where it matters, MathWorks tooling where the problem statement calls for it.

System Architecture & Data Flow

Client Tier (Role Based)

Android: ASHA capture,
offline sync, voice read-out
Web: reviewer queue +
district analytics dashboard

FastAPI Gateway

Patients, queue, review,
analytics, sync, audio
REST endpoints (PostgreSQL)

MATLAB Pipeline

QA → Enhance → Segment
→ Grade → Grad-CAM → Report

Simulink Capacity Model

Reviewer + bandwidth
sizing for 100k+/yr

FEASIBILITY AND VIABILITY

Feasibility: Core classical-CV pipeline runs end-to-end today with no trained model required; the trained severity classifier is tested on Messidor-2, a benchmark completely untouched during training — the honest, external standard most hackathon prototypes never check against.

▶ **Iteration Evidence (external Messidor-2 benchmark)**

Referable-DR sensitivity (target ≥ 0.90)



Final Deployed Model — Weighted Ensemble

Composition: I07 (0.376) + K10 (0.086) + K11 (0.538) — 3 independently seeded checkpoints
Sensitivity / Specificity: 0.7352 / 0.8508 · QWK: 0.5509
Effective sensitivity (AI + review): 0.8928 — near target
Review rate: 43.6% → ~73% reviewer utilization at 100k patients/yr (sustainable margin)

▶ **Key Risks — Identified & Addressed Through Testing**

Risk	Why it matters	Mitigation strategy
Cross-camera domain shift	Confirmed via external Messidor-2 testing	Ben Graham color normalisation + multi-source (APTOS+IDRiD) training
Rare-class (severe/PDR) data scarcity	Causes unstable, biased predictions	Class-weighted sampling + ordinal-aware loss + seed/checkpoint selection
Silent misdiagnosis risk	Any AI-alone system can be confidently wrong	Calibrated confidence gates mandatory human review, never a silent miss
District rollout planning	Unknown staffing/bandwidth needs	Simulink capacity model sizes reviewers & bandwidth for target volume

IMPACT AND BENEFITS

1

Social

- Early detection of preventable blindness for populations with least specialist access
- Native-language voice read-out of diagnosis + instructions removes the literacy/language barrier for ASHA-led rural delivery
- Turns ASHA workers & rural PHCs into real screening points

2

Economic

- Frees scarce ophthalmologist time for genuinely referable cases only, via automatic triage
- Avoids costlier late-stage treatment and lost productivity from preventable blindness
- Simulink capacity model lets a district right-size staffing/bandwidth investment, not guess

3

Environmental

- Fewer screening-only trips to distant clinics — lower travel cost, time, and carbon footprint
- Reduces load on already overstretched public hospital infrastructure
- Deployable on existing smartphone + low-cost fundus-lens hardware, no new infrastructure

101M

people living with diabetes in India.
ICMR-INDIAB, 2023

12.5%

of diabetics develop diabetic retinopathy.
Lancet Glob. Health, 2022

~12.6M

Indians estimated to have DR nationally.
derived: 101M × 12.5%

4.0%

have vision-threatening DR. (~4M people)
SMART-India study, 2022

Datasets, Methods & Sources

Category	Source	Key detail
Datasets	APTOS 2019 Blindness Detection	Kaggle — primary training set (~3,662 graded fundus images)
Datasets	IDRiD (Indian Diabetic Retinopathy Image Dataset)	idrid.grand-challenge.org — segmentation masks + disease grading, Indian population
Datasets	Messidor-2 images + adjudicated grades	ADCIS (images) + Kaggle google-brain/messidor2-dr-grades (Krause et al.) — external validation benchmark
Methods	ICDR Severity Scale	Wilkinson et al., “Proposed International Clinical DR and DME Disease Severity Scales,” Ophthalmology, 2003
Methods	Confidence calibration	Guo et al., “On Calibration of Modern Neural Networks,” ICML 2017
Methods	Cross-camera color normalisation	Graham, B., Kaggle Diabetic Retinopathy Detection — winning solution, 2015
Public health	101M Indians living with diabetes	Anjana et al., ICMR-INDIAB, Lancet Diabetes & Endocrinology, 2023
Public health	12.5% DR / 4.0% vision-threatening DR prevalence	SMART-India study, Lancet Global Health, 2022